Odd permutations at an attachment and the remaining kernel obstruction

Human Compression project: a component of the L019 recovery

For a set X, write $\mathrm{Sym}(X)$ for permutations supported on X, fixing every other point. We use actual supported subgroups throughout. The acquired leaf criterion says that for an anchored string C-marking $(h_0, \dots, h_{k-1})$, where all h_j with $j > 0$ fix a, adjoining a pure new transposition at a preserves IP exactly when

$$P_j \cap \mathrm{Sym}(X_{j-1}) \le P_{j-1} \quad (1 \le j \le k), \qquad P_j = \langle h_0, \dots, h_{j-1} \rangle, \quad X_j = a^{P_j}. \tag{1}$$

The first condition is automatic. We prove a uniform way to obtain these conditions from the IP of a larger, already given marking.

Lemma 1. *Let groups P, E have supports contained respectively in sets Ω, Y with $\Omega \cap Y = \{a\}$. Suppose E is transitive on Y, $|Y| \ge 2$, and put $X = a^P$. If $|X| \ge 2$, then*

$$\mathrm{Alt}(X \cup Y) \le \langle P, E \rangle.$$

If E contains an odd permutation, then $\mathrm{Sym}(X \cup Y) \le \langle P, E \rangle$.

Proof. For any $x \in X \setminus \{a\}$ and $y \in Y \setminus \{a\}$, choose $p \in P, e \in E$ with $a^p = x, a^e = y$. The supports of p, e intersect only at a. A direct evaluation in the right-action convention shows that $p^{-1}e^{-1}pe = (a\ y\ x)$, fixing every other point. Thus the generated group contains every three-cycle with a, one other point from X, and one from Y.

These cycles generate $\mathrm{Alt}(X \cup Y)$. To see this directly, fix $x_0 \in X \setminus \{a\}$ and $y_0 \in Y \setminus \{a\}$. For two distinct points $x, x' \in X \setminus \{a\}$, the product $(a\ y_0\ x)^{-1}(a\ y_0\ x')$ is the three-cycle $(a\ x\ x')$. The same argument with x_0 produces every three-cycle on a, y, y' for distinct $y, y' \in Y \setminus \{a\}$. Together with the mixed cycles, these are all three-cycles containing a. They generate the alternating group: any three-cycle avoiding a is a product of two such cycles, and every even permutation is a product of three-cycles. The argument also covers $|X| = |Y| = 2$. All displayed cycles are supported on $X \cup Y$.

An odd element of E is supported on $Y \subseteq X \cup Y$. Adjoining it to the alternating group gives the full supported symmetric group on that union, proving the final assertion. □

Theorem 2. *Suppose $(h_0, \dots, h_{k-1})$ is a submarking of a string C-marking and has endpoint anchor a. Let E be generated by a disjoint set of labels from the same C-marking. Suppose E is transitive on a set Y of size at least two and is supported there, while all the h_j are supported in Ω with $\Omega \cap Y = \{a\}$. Then every even part of a kernel in (1) is already in the preceding prefix:*

$$P_j \cap \mathrm{Alt}(X_{j-1}) \le P_{j-1}. \tag{2}$$

If E contains an odd permutation, all conditions (1) hold and the pure-leaf extension is C.

For arbitrary E, a failed condition (1) has precisely an index-two sign defect. In detail, with

$$Q_j = P_j \cap \mathrm{Sym}(X_{j-1}), \qquad K_j = P_{j-1} \cap \mathrm{Sym}(X_{j-1}),$$

one either has $Q_j = K_j$, or $K_j = Q_j \cap \mathrm{Alt}(X_{j-1})$ and $[Q_j : K_j] = 2$.

Proof. For $j = 1$, the supported symmetric group on $X_0 = \{a\}$ is trivial. If h_0 fixes a, every $X_j = \{a\}$ and all the conditions are trivial. Otherwise, for $j \geq 2$, we have $|X_{j-1}| \geq 2$. Apply Lemma 1 with $P = P_{j-1}$. It puts the supported $\mathrm{Alt}(X_{j-1})$, and also $\mathrm{Sym}(X_{j-1})$ when E has an odd element, inside $\langle P_{j-1}, E\rangle$. The original intersection property gives the exact bound

$$P_j \cap \langle P_{j-1}, E\rangle = P_{j-1},$$

since the label sets of E and the h_j are disjoint. This proves (2) and, in the odd case, every condition (1). The complete acquired leaf theorem then proves the C-property of the enlarged marking, including all its subgroup intersections.

Finally $K_j \leq Q_j$, and (2) gives $Q_j \cap \mathrm{Alt}(X_{j-1}) \leq K_j$. If K_j has an odd element it equals Q_j, because its even subgroup already contains the entire even part of Q_j. Otherwise it equals that even part. A strict inclusion is therefore possible only when Q_j has odd elements and K_j does not, in which case its index is exactly two. □

Corollary 3. *Let an internal perfect split of a transitive symmetric string C-marking have the form*

$$q = \rho_i = (a\ b)\beta, \quad L = \langle \rho_0, \ldots, \rho_{i-1}\rangle, \quad J = \langle q, \rho_{i+1}, \ldots, \rho_{r-1}\rangle.$$

Here the old left internal factor is trivial. If both L and J contain odd permutations, subdividing the crossing by replacing q with $(a\ c), (c\ b)\beta$ at a new point c gives a string C-marking of degree $n + 1$ and rank $r + 1$, generating S_{n+1}. The reversed assertion holds when the right internal factor is trivial.

Without the two oddness assumptions, at most one of the two leaf families in the acquired split criterion can fail, and every failure there is exactly the sign defect of Theorem 2.

Proof. The supported left group L is transitive on A. Since the split is internal, $|A| \geq 2$. The group J is supported and transitive on $\{a\} \cup B$: its tail is transitive on B, and q joins a to b. These two domains meet only at a. Apply Theorem 2 first to the old left wing in reversed order, with external group J, and then to the marking beginning with q and continuing rightwards, with external group L. These are exactly the two original lists in the repaired split criterion. Their endpoint-anchor hypotheses follow from the unique crossing and the actual wing transitivity. If both external groups have odd elements, both lists satisfy all the required tests. The complete split criterion gives the full C-property and string relations of the subdivided marking.

For its group, write $u = (a\ c), v = (c\ b)\beta$. Then $uvu = q$ on the old domain and fixes c. All other old marks are retained, so the new group contains the old S_n fixing c, together with u. Conjugating u by that S_n gives all transpositions $(x\ c)$; hence the group is S_{n+1}. The inserted involutions are distinct and nonidentity by the split criterion, so the rank increases by one.

Lastly $\langle L, J\rangle = S_n$, so at least one of these two actual groups contains an odd permutation. Its external use proves one whole leaf family. The other family has only the exact possible index-two defects already established; no assertion is made that these remaining defects are automatically absent in the high-rank range. □

Attribution and boundary. The extension and its intended application are those of Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2. The complete restriction-kernel criterion is the separately reviewed repair retained as an acquired premise, including its standard C-group end criterion attribution. The elementary commutator argument above supplies a new sufficient interface from the original IP, and identifies the only residual obstruction when the attaching group is even. No priority claim is made for the alternating-group generation identity itself.

This proves extension for the stated oddness condition. Existence of such a split for every high-rank marking, or elimination of the remaining sign defects by another method, has not been established. The reversible correspondence, exact base constants and whole L019 remain pending. No new census, Lean or compression measurement enters.